SAMPLE

History of Astronomy: Radio Astronomy History

Hearing the Invisible Universe

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Introduction

In 1932 a young radio engineer at Bell Laboratories, assigned to hunt the static that troubled transatlantic telephone calls, found something no one was looking for: a steady hiss from the sky itself, rising and falling once a day, drifting with the seasons, and strongest from the direction of the constellation Sagittarius, toward the center of the Milky Way. Karl Jansky had discovered, without intending to and with an instrument built for telephony, that the universe radiates radio waves, and the entire history this book recounts descends from his hiss. Radio astronomy is the field that hiss founded: the study of the sky at wavelengths the human eye cannot see, with instruments that look nothing like telescopes, and its history is one of the strangest and most productive chapters in all of science, a story in which telephone engineers, radar veterans, backyard amateurs, and Nobel laureates all take turns as protagonists.

The plan of the pages is the series' familiar one. The opening pages define the field, its physics, and its peculiar window: why the atmosphere happens to admit radio waves that it blocks everywhere else in the spectrum, and what that accident made observable. The middle chapters carry the history: Jansky's serendipity, Grote Reber's backyard dish, the wartime radar laboratories and the solar radio bursts their operators could not explain, the postwar explosion of discoveries, the hydrogen line that mapped the galaxy's spiral arms, the radio galaxies and quasars that broke the universe open, the pulsars, the cosmic microwave background, and the molecular clouds where astrochemistry was born. The later chapters describe the modern instruments, from the great dishes to the continent-spanning arrays, and the challenges, chiefly the crowding of the radio spectrum, that the field now faces.

A note on the field's character, because it explains the history's texture: radio astronomy has always been an engineering science, built by people who made their own instruments, funded for a while by telephone companies and militaries that wanted something else, and rewarded with a string of discoveries that the optical establishment, at first, flatly did not believe. The chapters that follow honor that texture, with the dates and the instruments given their due, and the controversies, of which the field has had several, told as history rather than smoothed away.

One orienting note on nomenclature and scope. The pages that follow speak of radio in the engineer's broad sense, wavelengths from millimeters to tens of meters, and of astronomy in the sense that matters, the measuring of celestial objects, so the term radio telescope covers dishes, arrays of dishes, and forests of simple antennas alike, and the term receiver means the whole amplifying chain that turns sky power into numbers. Historical dates are given to the year where the record is firm, and the small number of famously contested attributions, the credit disputes and the disputed detections, are flagged as such rather than smoothed, because the field's history is part of its science. The reader who wants the instruments can skim the chapter on observation; the reader who wants the revolutions can start at the discoveries; the book is built so that either order works.

A note on the window metaphor, because the book leans on it and the reader should know exactly what the metaphor carries. The atmosphere admits only two broad bands of the spectrum to the ground: the narrow band the eye uses, and the wide band the radios use, with the rest arriving blocked or mangled; the radio window is thus not a bonus but the second eye of a two-eyed science, the band that sees through the dust, hears the invisible, and works in daylight and storm; the account uses the window honestly throughout, the field being neither an adjunct of optical astronomy nor a rival but the other half of the ground-based whole, and the reader who holds the metaphor holds the book's frame.

A reader's map of the chapters, because the field's history braids several stories and the map prevents losing the threads. The instrument story runs from the rotating merry-go-round through the backyard dish to the planet spanning arrays; the discovery story runs from the steady hiss through the discrete sources to the exotic objects the window found; the technique story runs from single antennas to synthesis and correlation; and the chapters interleave the three, the account marking each thread's stations as they pass, the reader free to follow one thread across chapters or take the braid whole, the book being written for both habits.

Chapter 1: The Discovery of Radio Astronomy

The discovery of cosmic radio waves was prepared by two unconnected developments: the proof that such waves exist at all, and the growth of a communications industry that needed to understand interference. The theoretical foundation was laid in the 1860s and 1880s, when Maxwell's equations predicted electromagnetic radiation and Hertz generated and detected it in the laboratory; the practical foundation was the transatlantic radio telephony of the 1920s, whose operators found their circuits washed with static of unknown origin. Bell Telephone Laboratories,

needing to characterize that static to build better links, assigned the problem to Karl Jansky, a young engineer who in 1929 built a rotating antenna array, a bristle-backed structure his colleagues nicknamed the merry-go-round, tuned to a wavelength of about fifteen meters, to record the noise from every direction around the horizon.

Jansky's data sorted itself into three sources, as he reported in his papers of 1932 and 1933: local thunderstorms, sharp crashes of static; distant tropical storms, a general crackle; and a third component, a faint steady hiss that peaked once every sidereal day, four minutes earlier each solar day, following the stars rather than the Sun. The repetition tied the hiss to the sky, and its point of origin, tracking through the constellations, settled in the direction of Sagittarius, where astronomers already suspected the galaxy's center lay. Jansky, careful with his language, concluded that the static was of extraterrestrial origin, probably thermal radiation from the Milky Way's disk; the newspapers of May 1933 carried the story, some of them more excited than the stolid Bell system considered proper, and the engineer gave interviews and a lecture, hoping to pursue the phenomenon with a better antenna.

The company had other plans. Jansky was reassigned to work on ordinary telephone problems, and his request for a dedicated radio telescope was declined; the sky's hiss, having been characterized to the telephone company's satisfaction, was filed away as an engineering curiosity. He died in 1950, at the age of forty-four, just before the field he had founded entered its great period; the unit of radio flux, the jansky, now carries his name, and the field's flagship journal and its prizes wear it too, an unusual weight of commemoration for a man the industry kept from his own discovery.

The discovery's second act belonged to an amateur. Grote Reber, a radio engineer in Wheaton, Illinois, read Jansky's papers, asked for the field and was told there was none, and in 1937 built in his own backyard a parabolic dish, nine and a half meters across, the first purpose-built radio telescope in the world, and spent the following years surveying the sky with it, working nights around his day job. His maps, published in the early 1940s, confirmed the galactic hiss, found the sky brighter at the longer meter wavelengths than theory could comfortably explain, and located the intense sources in Cassiopeia and Cygnus that the postwar generation would chase to their revolutionary identities. Between them, one professional ignored and one amateur funded, the field's discovery decade established the pattern its whole history would follow: the instruments would be improvised, the funding accidental, and the sky more interesting than anyone budgeted for.

The near-miss file belongs in the discovery chapter, because the hiss was almost found earlier, twice. In the first years of the twentieth century, radio pioneers on both sides of the Atlantic reported hearing whistling and signals of unknown origin in their receivers, and some of what they logged was probably nature's own, atmospheric and cosmic, mixed beyond recovery with the infant technology's own faults; and in the 1920s, researchers investigating transatlantic radio propagation measured sky noise as a nuisance parameter, without asking where it came from. The lesson the historians draw is the field's founding one: discovery required a question, and the question required a business reason, since it was the telephone company's need to characterize interference, not curiosity about the sky, that finally put a calibrated rotating antenna under a patient engineer's hands. Serendipity, in this history, is real but overrated; preparation plus payroll is the winning formula, and the field's whole first decade is the proof.

The prehistory's paradox deserves the discovery chapter's first addition: the waves were proven in the laboratory decades before anyone looked for them in the sky. The generation of radio waves in the late eighties of the nineteenth century confirmed the theory that light and radio are one phenomenon, and the obvious question, whether the sun and the stars broadcast as well as shine, was asked aloud by the era's physicists and even attempted by its inventors; the attempts failed for honest reasons, the receivers being crude by factors of millions; and the chapter's honest judgment: the cosmic radio sky was predicted early, sought early, and found late, the delay being an instrument's biography rather than an idea's, the field waiting seventy years for ears sensitive enough.

The backyard sequel takes the second addition, because between the accidental discovery and the war's mobilization stands one man with a dish in his garden. The young engineer who read the discovery's papers and asked for the problem was refused by the institutions of the era, the depression's budgets closing the professional doors; he built his own instrument instead, a paraboloid of his own design in his Illinois backyard, the thirties' only radio telescope in the world, its builder surveying the sky on his own money in his own nights; the sequel's outcome: the first radio map of the galaxy and the first source catalogs, the field's second founder being an amateur by circumstance and a professional by method, and the account keeping his years alone as the field's proof that its history is a history of stubbornness as much as physics.

The wartime radar people deserve the discovery story a closing card, because the field's second generation was trained by the war. The mobilization put thousands of engineers onto receivers and antennas, the chain of radar

stations and the portable sets teaching a cohort to build sensitive equipment and to distrust their own noise, and when the war ended the veterans turned surplus hardware skyward, the ripple of the era's skills spreading through the late 1940s like a spring thaw; the amateurs and the radar men together turned the hiss into a science, the field's founding population being a generation the war had happened to educate. The card is the history's hinge: a peacetime discipline was assembled from a wartime one, the receiver's sensitivity being the shared organ, the veterans' surplus dishes and the amateurs' backyards forming one community, the radio sky having been opened twice, once by accident and once by demobilization.